

Digitap.ai Statement Upload UI API Specifications

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A jumpstart guide to boosting your Business

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1 Process Flow

1.1 Description

This documentation is designed to give an overview of the Digitap BankData UI related API(s) - how it works, what are the request-response and error scenarios for each API. Detailed descriptions of the same will be available in the next sections.

1.2 Functionalities

There are two features supported under Digitap BankData Pdf Upload:

A. Single Account PDF Upload

This feature allows users to upload bank statements for a single bank account. Users can upload multiple PDFs, but all must pertain to the same account.

Example Case:

- **Bank Name:** SBI Bank
- **Account Number:** XXXXXXXX88657
- **Upload Details:** All uploaded bank statements must belong to the above account number from SBI Bank.

B. Multiple Account PDF Upload

This feature supports uploading bank statements for multiple accounts, whether they belong to the same bank or different banks. Users can upload multiple PDFs for various accounts.

Example Cases:

Case 1 : Multiple accounts from the same Bank - If a user has 2 accounts in the same bank, then we support the upload of PDFs of both these accounts.

- **Bank Name:** HDFC Bank
- **Accounts:**
 - **Account Number_1:** XXXXXXXX990078
 - **Number of PDFs:** 2
 - **Account Number_2:** XXXXXXXX784590
 - **Number of PDFs:** 3

Case 2: Multiple accounts from different Banks - If a user has multiple accounts in multiple banks, we allow the user

- **Bank_1:** Axis Bank
- **Accounts:**
 - **Account Number_1:** XXXXXXXX675489
 - **Number of PDFs:** 3

 - **Account Number_2:** XXXXXXXX554380
 - **Number of PDFs:** 3

- **Bank_2:** ICICI Bank
- **Accounts:**
 - **Account Number_1:** XXXXXXXX339489
 - **Number of PDFs:** 3

 - **Account Number_2:** XXXXXXXX228380
 - **Number of PDFs:** 3

NOTE : The Single Account PDF Upload feature is for uploading statements of a single account, while the Multiple Account PDF Upload feature is for uploading statements of multiple accounts from one or more banks. The restrictions on uploading multiple account bank statements are configurable as per the client requirements. Please refer the following table :

Restrictions Table:

Key Name	Significance
no_of_institution(Banks)	This restriction is based on how many different institutions (banks) are allowed for bank statement parsing in a transaction.
no_of_accounts_per_institution	This restriction is based on how many accounts per institution are allowed for bank statement parsing.
no_of_statement_per_account	This restriction is based on how many statements are allowed to parse for an account belonging to the

	respective institution.
no_of_statement_per_request	This restriction is based on how many statements we are allowed through a request.

1.3 How Does It Work?

Following steps will describe the process flow of the Digitap Bank Data UI:

- **Step 1:** The Client starts by sending a request to the Generate URL API, this API will verify the client's credentials and generate a URL which the client can share with the customer for completing the further process or can be used in the app inside the iFrame.
- **Step 2:** Based on the "**destination**" parameter given in the above API request, the end user will be redirected to the appropriate Digitap UI Page. If the destination value is given as "**statementupload**" then the end user will land on the Digitap PDF Upload Page. After the customer completes uploading the PDF of the selected institution and finishes the upload then the customer is then redirected back to the URL specified by the client under the parameter("return_url") in Generate URL API request.
- **Step 3:** Now the client can invoke the Status Check API to know the status of the transaction. Clients will receive the response in the form of JSON in which transaction status will be mentioned, which can be successful or failure. If the client receives the status as **ReportGenerated** then the report for the customer will be ready. Alternatively, clients can also know the final processing completion status when the callback is received in the txn_completed_cburl webhook.
- **Step 4:** If Step 3 is successfully done, then the client can invoke the Retrieve Report API to get the reports.

2 API Requests & Authentication

2.1 API Requests

API requests need to have the following properties:

- Content-Type will be "application/json" (i.e. API requests have header "content-type": "application/json")

2.2 Authentication

We offer two distinct authentication processes to cater to different needs of client:

- Signature based authentication
- Header based authentication.

1. Signature-Based Authentication:

Our signature-based authentication process ensures the integrity and authenticity of requests by utilizing digital signatures. With this method, requests are signed using a public key, and the receiving system verifies the signature using the corresponding private key. This approach provides a secure means of validating the authenticity of requests.

2. Header-Based Authentication:

In addition to signature-based authentication, we also support header-based authentication. This method involves including credentials (such as a client Id and secret) in the request headers. The server or system validates these credentials to grant access. Basic authentication offers a simpler authentication mechanism that is widely supported across various systems.

By offering both signature-based authentication and header-based authentication, we provide flexibility for our clients to choose the method that best aligns with their specific requirements and capabilities. This allows for seamless integration into a variety of applications and systems while maintaining the necessary security measures.

2.2.1 Signature-Based Authentication Method:

Signature-based authentication is a method that ensures the integrity and authenticity of requests by using digital signatures. Signed requests made from the client should be signed with the key shared with the client (*it will be different based upon the UAT/ PROD environment*).

The signature of a request is calculated as follows:

1. Construct the request payload that needs to be sent to the server.
2. Stringify the JSON request payload.
3. Generate a SHA256 digest of the stringified JSON.
4. Base 16 encode generated digest
5. Encrypt the digest using the *Digitap-provided Key*
6. Base 16 encode the encrypted digest.
7. Pass the above output in the signature key of Input JSON.

Pseudo-code for the same:

```
signature = base16(encrypt(base16(sha256(stringify(json_payload))));
```

Sample Code:

Refer to the GitHub Repository - https://github.com/DigiTapAi/digitap_bank_data_api_public for sample code for Signature Generation across different programming languages

Note: The public key should be stored in a secure location with restricted access. Consider using a hardware security module (HSM) or other secure key storage mechanisms.

2.2.2 Header-Based Authentication Method:

Header-Based Authentication is an authentication scheme used to authenticate API requests. It requires the client to include a username(Client ID) and password(Client secret) in the request headers using basic authentication.

To authenticate using Header-based Authorization, the client needs to include an **Authorization** header in the following format:

Authorization: Basic <credentials>

In the above format, <credentials> represents the Base64-encoded username and password combination. The username and password are concatenated with a colon (":") in the format "client_id:client_secret", and then Base64-encoded.

For example, if the Client ID is "1234652772" and the client secret is "absdgwhgwhg21g", the credentials would be "1234652772:absdgwhgwhg21g". The Base64-encoded value of "client_id:client_secret" is "MTIzNDY1Mjc3MjphYnNkZ3doZ3doZzlxZW==".

3 API Responses

3.1 API Response Properties

API responses will have the following properties:

- Content-Type will be application/json
- HTTP status code 200 will indicate that the call succeeded
- Non-200 status codes will be returned if something went wrong
- Appropriate status codes will be returned according to the list mentioned in the next sections.

3.2 API Response Structure

When the API is processed by Digitap, the API returns a JSON response with a key “**status**” indicating that the API is successful or an error.

```
{
  "status": "success",
  ...
}
```

Or

```
{
  "status": "error",
  ...
}
```

3.3 Error Responses

The following sample error response shows the structure of response elements common to all error responses.

NOTE: In case of error response HTTP code will be other than 200.

```
{
  "status": "error",
  "code": "InvalidInstitution",
  "msg": "Client has sent an Institution Id that is not supported by Digitap"
}
```

4 Generate URL API

4.1 URL

- DEMO Env - <https://svcdemo.digitap.work/bank-data/generateurl>

Note: DNS will be shared separately for the PROD environment; senders just need to add the described path or parameter.

4.2 Accepted Method

POST

4.3 Input JSON Structure

As specified above in Authentication, based on the method chosen by the client the payload will differ and the same is mentioned below:

4.3.1 For Signature-Based Authentication Method

The method includes sending payload and signature in the JSON request, the signature should be generated dynamically based on the the payload that is to be sent and the JSON data should be enclosed as shown below:

- **Sample Payload(For Single Account Support)**

Request Body: {

```
"payload":{
    "client_name": "client_xxx",
    "client_ref_num" : "abcd123",
    "txn_completed_cburl": "http://client-given-callback-url",
    "start_month": "2020-01",
    "end_month": "2020-02",
    "institution_id": "1",
    "destination": "statementupload",
    "return_url": "http://api.clientxxx.com/digitap/?transID=%s&status=%s",
    "acceptance_policy" : "atLeastOneTransactionInRange",
    "relaxation_days" : 0,
},
"signature": "xxxx"
}
```

- **Sample Payload (For Multiple Account Support)**

Request Body: {

```
"payload":{
    "client_name": "client_xxx",
    "multi_account_support_required": "1",
    "client_ref_num" : "abcd123",
    "txn_completed_cburl": "http://client-given-callback-url",
    "start_month": "2020-01",
    "end_month": "2020-02",
    "institution_id": "1",
    "destination": "statementupload",
    "return_url": "http://api.clientxxx.com/digitap/?transID=%s&status=%s",
    "acceptance_policy" : "atLeastOneTransactionInRange",
    "relaxation_days" : 0,
}
```

```

},
"signature": "xxxx"
}

```

4.3.2 For Header-Based Authentication

The method included sending the Authorization Key in header with Value as base64 encoded ClientID and ClientSecret provided by Digitap, as shown below:

Headers: Authorization: Basic "base64-Encoded-ClientID:ClientSecret"

● Sample Payload(For Single Account Support)

Request Body: {

```

"client_ref_num" : "abcd123",
"txn_completed_cburl": "http://client-given-callback-url",
"start_month": "2020-01",
"end_month": "2020-02",
"institution_id": "1",
"destination": "statementupload",
"return_url": "http://api.clientxxx.com/digitap/?transID=%s&status=%s",
"acceptance_policy" : "atLeastOneTransactionInRange",
"relaxation_days" : 0

```

}

● Sample Payload (For Multiple Account Support)

Request Body: {

```

"client_ref_num" : "abcd123",
"multi_account_support_required": "1",
"txn_completed_cburl": "http://client-given-callback-url",
"start_month": "2020-01",
"end_month": "2020-02",
"institution_id": "1",
"destination": "statementupload",
"return_url": "http://api.clientxxx.com/digitap/?transID=%s&status=%s",
"acceptance_policy" : "atLeastOneTransactionInRange",
"relaxation_days" : 0

```

}

4.4 Payload Elements

Element Name	Mandatory(M) / Optional(O)	Description
--------------	-------------------------------	-------------

client_name	M	Name of the Client given by the Digitap. Mandatory for Signature Based Authentication and Not Required for Header based Authentication.
destination	O	Destination will be “ statementupload ” for PDF Upload UI. If not specified, then the default destination value will be taken based on the client configuration.
client_ref_num	M	Client-specified identifier to uniquely identify this transaction. Alpha-numeric, or dash (-), maximum 50 characters.
txn_completed_cburl	M	Digitap will send an HTTP POST request to this URL once the transaction is complete. Please refer to the “ Appendix A - Transaction Complete CallBack URL ” Section for more details.
start_month, end_month	O	Condition for the date range for which data should be fetched. Must be a valid year-month in YYYY-MM format. If not provided then the default value will be taken (<i>the client-specific configuration</i>). Note: However, if you'd prefer that users be allowed to upload bank statements from any time period without restrictions, we can enable a configuration on our end to bypass date range validations for uploaded statements.
institution_id	O	Institution or Bank ID given by Digitap Institution List API . • If specified, the user would be limited to a specified institution. • If not specified, users can choose from supported Institutions or banks
return_url	O	Send the End User to this URL after completing interaction with Digitap. Placeholders can be inserted in the URL to get a few data during the redirection. It supports two or three placeholders in the URL 1. Two placeholders case - will be replaced by the txn_id and true/false indicating Transaction success/failure in order 2. Three placeholders case - will be replaced by the request_id, txn_id and true/false indicating Transaction success/failure in order User will be redirected to the return URL once the transaction is complete, which will be one of the

		following cases: • When the User Clicked Cancel Button in the Home Page • When the User Clicked Take me Back button in the Error Page • Once PDF Parsing completed (Success) If return_url is not given by the client, no redirection would happen. If the number of placeholders is not 2 or 3, then no replacement would occur.
employer_details	O	Optional parameter to detect the salary transactions. Please refer to Appendix B - Employer Details Parameter Specification for more details.
acceptance_policy	O	Condition under which the transaction data should be accepted and report generated. Can be one of the below possible values: atLeastOneTransactionInRange : proceed if the user has at least one transaction in the expected date range. atLeastOneTransactionPerMonthInRange : proceed if the user has at least one transaction per month in the expected date range. exactStatementRange : proceed if the statement date range from all uploaded statements spans all the valid dates in the expected date range. Please refer to Appendix E - A note on Acceptance Policies for more details.
relaxation_days	O	Buffer days can be removed from end_date while considering the acceptance_policy and accepting the statement. <i>Work only if the destination is statetmentupload and acceptance_policy is exactStatementRange.</i>
multi_account_support_required	O	In case of multi account support this key is mandatory.

4.5 Success Response

When the API is successful, the following sample JSON response would be returned along with the HTTP status code as 200. The elements of the response are described as follows:

```
{
  "status": "success",
  "url": "http://digitap-bs-frontend-url",
  "expires": "2020-04-10T07:01:35.054",
  "request_id": "11dad12"
}
```

Elements	Description
status	This will denote the status of the call for Generate URL API, either success or error
url	This will have the digitap Bank data UI URL on which customer will do interactions
expires	It will denote the expiry of the URL.
request_id	Digitap generated a unique request id for each call of the Generate URL API.

4.6 Error Response

Below is the sample error response with the element “**code**” denoting the error code and “**msg**” describing the error.

```
{
  "status": "error",
  "code": "InvalidInstitution",
  "msg": "Client has sent an Institution Id that is not supported by Digitap"
}
```

4.7 List of API Error Codes

For the status as **error**, following are the possible values for the element **code**

Code	Description
ClientNotConfigured	The configuration is not yet created for the client.

Code	Description
	Please contact our support team to set up and ensure smooth access to our services.
AccessDenied	The client is sent invalid credentials or is not permitted to access a URL. Please contact our support team.
MalformedJson	Client has sent malformed JSON for the API
InvalidInstitution	Client has sent an Institution Id that is not supported by Digitap
InstitutionCurrentlyNotSupported	Institution Not supported by Digitap (due to internal or external reasons)
SignatureDoesNotMatch	The request signature we calculated does not match the signature you provided
InvalidEncryption	We are unable to decrypt the signature using the private key
NotSignedUp	Client has not signed up for the Digitap Bank data Service
DateRangeTooLarge	The date range specified by the Client exceeds the contractually permitted limit
InvalidStmtStartDate	Invalid Statement Start Date
InvalidStmtEndDate	Invalid Statement End Date
InvalidDestination	Invalid value given for destination parameter.
InvalidClientRefNum	Client has sent an unsupported client reference number
InvalidTxnCompletedCBaseUrl	Invalid transaction completed callback url
InvalidAcceptancePolicy	Invalid Acceptance policy
InvalidStmtStartDate	Invalid Statement Start Date
InvalidStmtEndDate	Invalid Statement End Date
RequiredPayload	Payload is required but missing
RequiredSignature	signature is required but missing
RequiredClientName	Client Name is Invalid/Missing

Code	Description
InternalError	We encountered an internal error. Please try again
StartDateNotSupported	Start date cannot be older than 1 year
EmployerDetailsError	Employer details provided are either incomplete or improper.
InstitutionDestinationError	Given institution ID is not compatible with the given destination (Raised when Institution ID is passed with Choice Dest.)
RateLimited	Our system has detected more traffic and has rate-limited your IP.

4.8 List of Statement Upload Error Codes

This section lists the possible validation errors that can occur during statement upload, divided into two categories:

- **Pre-checks:** Errors that occur before parsing the statement.
- **Post-checks:** Errors that occur after parsing, during data validation.

4.8.1 Pre-checks

These validations happen before statement parsing, based on file structure, metadata, and basic rules.

Code	Description
File editor blacklisted	The file editor is blacklisted.
StatementIsScannedImage	Statement is scanned image
Invalid pdf type	Unable to detect the Bank Statement format for the selected bank. Either the statement belongs to a different bank or an unsupported format.
Tamper error	Internal Error with low confidence score.
No file password	No password provided for password

	protected file
Invalid file password	Wrong password provided to open a password protected file
Multiple accounts found	More than 1 account found in the bank statement.
Mpassbook date range error	You are trying to upload the incomplete bank statement PDF, please download the bank statement for the entire range and try again.
PDF extraction error	The uploaded statement does not seem to be an original bank statement. Please get the original bank statement and try again.
Txn date range	No bank transactions in the expected date range
Internal error	Internal error occurred. Please try again.
Missing transaction error	Statements are missing some transactions. Please download the complete statement from the bank website.
Balance column missing error	Balance column is not present in the statement. Please upload the statement where the balance column is present.
Different accounts found	Uploaded statement account details are different from previous statements. Please upload the statement of the same account. (Error Code: 086)
Insufficient customer data	Insufficient customer data. Please upload the statement where customer data is present
File too large	Client tried to upload a file whose size is larger than the maximum allowed
Invalid bank account type	The account type of the statement is not supported.
Unsupported PDF	NULL
PDFScoreLessConfidenceError	The uploaded statement does not seem to be an original bank statement. Please get the original bank statement and try again.
StmtTemplateNotSupported	The statement template is not supported by Digitap.

UserCancelled	User Cancelled the Transaction
TxnExpired	The requested transaction expired. (Error Code: 088)
AutoDetectionError	We could not detect the bank from the uploaded pdf. Please check if it is a valid bank pdf and try again.

4.8.2 Post-checks

These validations happen after parsing, when the system analyzes extracted transaction data for completeness and accuracy.

Code	Description
IncompleteMonths	Following are some possible scenarios due to failure in validation when amount validated across multiple pdfs • Same transaction but with different narration present in any of the uploaded pdf. • Missing transactions due to bank restriction on number of transactions allowed in single pdf. • Duplicate transactions present in pdf. • Transaction with empty narration present in pdf. • Actual missing transaction. For instance, Transactions for a date range present in pdf and balance are not matching across pdfs.
TxnDateRange	Following are some possible scenarios- • This occurs when the date range in the Bank Statement matches with the date range given by the user as per the acceptance policy but there are no transactions in the Bank Statement.
TxnPerMonthError	Following are some possible scenarios - • This occurs when the date range in the Bank Statement matches with the date range given by the user as per the acceptance

	policy but there are no transactions in the Bank Statement.
TxnExactDateRange (Exact Statement Date Range)	Following are some possible scenarios - • This occurs when the date range in the Bank Statement matches with the date range given by the user as per the acceptance policy but there are no transactions in the Bank Statement.
Tamper error	Internal Error with low confidence score.
Txn date range	No bank transactions in the expected date range
Internal error	Internal error occurred. Please try again.
TxnExpired	The requested transaction expired. (Error Code: 088)

5 Status Check API

5.1 URL

- DEMO Env - <https://svcdemo.digitap.work/bank-data/statuscheck>

Note: DNS will be shared separately for the PROD environment; senders just need to add the described path or parameter.

5.2 Method Accepted

POST

5.3 Input JSON Structure

As specified above in Authentication, based on the method chosen by the client the payload will differ and the same is mentioned below:

5.3.1 For Signature-Based Authentication Method

The method includes sending payload and signature in the JSON request, the signature should be generated dynamically based on the the payload that is to be sent and the JSON data should be enclosed as shown below:

```
Request Body: {
    "payload": {
        "client_name": "clientxxx",
        "request_id": "yyyyy"
    }
    "signature": ""
}
```

5.3.2 For Header-Based Authentication

The method included sending the Authorization Key in header with Value as base64 encoded ClientID and ClientSecret provided by Digitap, as shown below:

Headers: Authorization: Basic "base64-Encoded-ClientID:ClientSecret"

```
Request Body: {
    "request_id": "yyyyy"
}
```

5.4 Payload Elements

Element Name	Mandatory(M) / Optional(O)	Description
client_name	M	Name of the Client given by the Digitap. Mandatory for Signature Based Authentication and Not Required for Header based Authentication.
request_id	M	Digitap provided unique request_id (returned during the Generate URL API response)

5.5 Success Response

The successful sample response and the elements of the response are described as follows:

Case 1: Transaction is successfully completed

```
{
```

```

    'status': 'success',
    'request_id': '223344',
    'txn_status': [
        {'code': 'ReportGenerated',
         'status': 'Success',
         'msg': 'The report has been successfully generated for the transaction',
         'txn_id': 'daf4292'}
    ]
}

```

Case 2: Transaction expired

```

{
  "status": "success",
  "request_id": "191014",
  "txn_status": [
    {
      "code": "TxnExpired",
      "status": "Failure",
      "msg": "Requested transaction expired. (Error Code: 088)",
      "txn_id": "da114fc9",
    }
  ]
}

```

Elements	Description
status	status of the api call success or error
request_id	Digitap provided unique request_id (<i>returned during the Generate URL API response</i>)
txn_status	List of all the transactions generated for the given request_id. Each element will represent one transaction having txn_id , status , code , and msg as keys. Below are the description for each keys:

Sub Elements	Description
status	Possible values are Success , Failure , Error or InProgress
code	Code denoting the specific scenario list is provided in the next section.

msg	Brief message description for either success or error response
txn_id	Digitap generated unique transaction id

Below are the possible values for the element “code” in case of status as “success”

Code	Description
TxnInitiated	Generated txn url and transaction id verified and initiated txn
TxnProcessing	Transaction is under process. Please try after sometime
ReportGenerated	The report has been successfully generated for the transaction
TxnNotCompleted	The Digitap Transaction referred by Client has not been Completed
TxnDateRange	No bank transactions in the expected date range
MultipleValidBankAccount	Multiple valid Accounts found for the given customer id. Please use a different bank account or try to upload a bank statement.
NoValidAccType	No valid bank account found for the given customer id. Please use a different bank account
StatementDateRange	Statements not uploaded for all days in expected date range
NoFileUploaded	No file uploaded successfully for the transaction
IncompleteMonths	Statements contain incomplete months data. Please upload data for complete months.
UserCancelled	The user cancelled the Transaction.
Txn_expired	The requested transaction expired.

5.6 API Error Response

Below is the sample error response with element **code** denoting the error code and **msg** describing the error.

```
{
  "status": "error",
  "code": "NoActiveTxn",
  "msg": "We could not find an active transaction corresponding to this txn_id"
}
```

5.7 List of API Error Codes

For the status as **error**, following are the possible values for the element **code**

Code	Description
ClientNotConfigured	The configuration is not yet created for the client. Please contact our support team to set up and ensure smooth access to our services.
AccessDenied	The client is sent invalid credentials or is not permitted to access a URL. Please contact our support team.
InternalError	We encountered an internal error. Please try again
InvalidEncryption	We are able to decrypt the signature using the private key
SignatureDoesNotMatch	The request signature we calculated does not match the signature you provided
RequiredPayload	Payload is required but missing
RequiredSignature	signature is required but missing
RequiredClientName	Client Name is Invalid/Missing
TxnNotFound	We could not find the Digitap Transaction referred to by the client
NoActiveTxn	We could not find an active transaction corresponding to this transaction Id
RequiredRequestId	Request id is required but missing
RateLimited	Our system has detected more traffic and has rate-limited your IP.

6 Retrieve Report API

6.1 URL

- DEMO Env - <https://svcdemo.digitap.work/bank-data/retrievereport>

Note: DNS will be shared separately for the PROD environment, senders just need to add the described path or parameter.

6.2 Method Accepted

POST

6.3 Input JSON Structure

As specified above in Authentication, based on the method chosen by the client the payload will differ and the same is mentioned below:

6.3.1 For Signature-Based Authentication Method

The method includes sending payload and signature in the JSON request, the signature should be generated dynamically based on the the payload that is to be sent and the JSON data should be enclosed as shown below :

Request Body: {
 "payload": {
 "client_name": "clientxxx",
 "txn_id": "zzzz",
 "report_type": "json",
 "report_subtype": "type1"
 }
 "signature": "xxx"
}

6.3.2 For Header-Based Authentication Method

The method included sending the Authorization Key in header with Value as base64 encoded ClientID and ClientSecret provided by Digitap, as shown below:

Headers: Authorization: Basic "base64-Encoded-ClientID:ClientSecret"


```
Request Body:{
    "txn_id" : "zzzz",
    "report_type":"json",
    "report_subtype":"type1" ,
}
```

6.4 Payload Elements

Element Name	Mandatory(M) / Optional(O)	Description
client_name	M	Name of the Client given by the Digitap. Mandatory for Signature Based Authentication and Not Required for Header based Authentication.
txn_id	M	Digitap generated unique transaction id <i>(returned in the txnstatus call or provided in txn_completed_cburl)</i>
report_type	O	Describes the format of the report either "json" or "xlsx" . If not, then the default value can be taken according to SLA.
report_subtype	O	Describing the type of report, possible values for json are "type1", "type2", "type3" and for xlsx is "type3". If not, then the default value can be taken according to SLA.
request_type	O	The "request_type" parameter is Not Required when requesting reports generated by Digitap. However, it is Required when fetching the original statements uploaded by the user. There are two values that can be passed for the "request_type" parameter: • "original_statement": This value will retrieve all the original statements, compress them into a zip file, and provide a presigned URL in the response. • "original_statement_password": This value will retrieve all the passwords and send them in the response. Note: Fetching Original Statement is only allowed in case of netbanking and statement

		upload & Client Needs to Sign Up for this service.
base64	O	The parameter base64 will be used to encode the Original Statement URL into base64 format. Clients have to decode this at their end using base64 decoder.

NOTE: Parameter *txn_id* can be found from one of the following ways:

- *txn_id* returned during the Status Check API. The *txn_id* corresponding to the ReportGenerated
- *txn_id* returned in the return url (placeholder provided for redirection of UI)
- *txn_id* returned in the response posted on the *txn_completed_cburl*
- For more detailed information of *report_type* and *report_subtype* check Appendix [C - Report Types and Subtypes](#)

6.5 Sample API Response

The response body will be an XLSX/JSON report. Content type will be as follows:

Report Type	Content Type
json	application/json
xlsx	application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

6.6 API Error Response

```
{
  "status": "error",
  "code": "TxnNotFound",
  "message": "We could not find the Digitap Transaction referred by the client"
}
```

6.7 List of API Error Codes

Code	Description
------	-------------

ClientNotConfigured	The configuration is not yet created for the client. Please contact our support team to set up and ensure smooth access to our services.
AccessDenied	The client is sent invalid credentials or is not permitted to access a URL. Please contact our support team.
InternalError	We encountered an internal error. Please try again
InvalidEncryption	We are enabled to decrypt the signature using the private key
SignatureDoesNotMatch	The request signature we calculated does not match the signature you provided
RequiredPayload	Payload is required but missing
RequiredSignature	signature is required but missing
RequiredClientName	Client Name is Invalid/Missing
TxnNotCompleted	The Digitap Transaction referred by Client has not been Completed.
TxnNotFound	We could not find the Digitap Transaction referred to by the client.
InvalidReportType	Invalid report type given or client is not allowed for the report type.
MaxReportPullAllowedError	The max allowed retrieve report count is exceeded.
AccessDenied	Client is not permitted to access this URL
OriginalStatementNotFound	We could not find any statements related to the transaction id provided.
ReportDeletionError	We could not find an active transaction corresponding to this transaction Id
OriginalStatementPasswordNotFound	No Passwords were found for the provided transaction ID.
RateLimited	Our system has detected more traffic and has rate-limited your IP.

7 Institution List API

7.1 URL

- DEMO Env - <https://svcdemo.digitap.work/bank-data/institutions>

Note: DNS will be shared separately for the PROD environment; senders just need to add the described path or parameter.

7.2 Method Accepted

GET & POST

7.3 Input JSON Structure

As specified above in Authentication, based on the method chosen by the client the payload will differ and the same is mentioned below:

7.3.1 For Signature-Based Authentication Method

The method includes sending payload and signature in the JSON request, the signature should be generated dynamically based on the the payload that is to be sent and the JSON data should be enclosed as shown below:

GET: Send the below payload as query_params

POST: Send the payload as JSON Data

```
Params / Request Body: {  
    "payload": {  
        "client_name": "clientxxx",  
        "type": "type"  
    }  
    "signature": "xxx"  
}
```

7.3.2 For Header-Based Authentication Method

The method included sending the Authorization Key in header with Value as base64 encoded ClientID and ClientSecret provided by Digitap, as shown below:

GET: Send **type** as a query_param

POST: Send the below payload as JSON Data

Headers: Authorization: Basic "base64-Encoded-ClientID:ClientSecret"
Request Body: { "type": "type" }

7.4 Payload Elements

Element Name	Mandatory(M) / Optional(O)	Description
client_name	M	Name of the Client given by the Digitap. Mandatory for Signature Based Authentication and Not Required for Header based Authentication.
type	M	For which API service Institution list is required, for statement upload it will be "Statement"

7.5 Sample API Response

```
{
  "status": "success",
  "data": [
    {
      "id": 1,
      "name": "Institution name",
      "inst_type": "Institution type"
    },
    {
      "id": 2,
      "name": "Institution name",
      "inst_type": "Institution type"
    }
  ]
}
```

Elements	Description
status	status of the api call success or error
data	List of all the institutions supported by Digitap Each element will have the id, name, type

Sub Elements	Description
id	ID given by the Digitap for the Institution
name	Name of the Institution
inst_type	Type of the Institution. Possible values are bank or other

7.6 API Error Response

Sample Error Response:

```
{
  "status": "error",
  "code": "InvalidType",
  "message": "Specified Type is invalid"
}
```

7.7 List of API Error Codes

Code	Description
ClientNotConfigured	The configuration is not yet created for the client. Please contact our support team to set up and ensure smooth access to our services.
AccessDenied	The client is sent invalid credentials or is not permitted to access a URL. Please contact our support team.
InternalError	We encountered an internal error. Please try again
InvalidEncryption	We are enabled to decrypt the signature using the private key
SignatureDoesNotMatch	The request signature we calculated does not match the signature you provided
RequiredPayload	Payload is required but missing
RequiredSignature	signature is required but missing
RequiredClientName	Client Name is Invalid/Missing

InvalidType	Specified Type is invalid
RateLimited	Our system has detected more traffic and has rate-limited your IP.

Appendix

A - Transaction Complete Callback URL

The **txn_completed_cburl** parameter given in the Generate URL API should be the RESTFUL API URL capable of receiving the callbacks. The same will be used to post the response by Digitap once the transaction is completed. The response will be posted in case of either successful or failure of the transaction.

The success and failure of a transaction is defined in the following points:

- When User Pressed Cancel in Home Page (Failure)
- When a User Pressed Take me back in the Error Page (Failure)
- Once PDF Parsing completed & Report Generation Succeeded(Success)

Digitap will attach the following header while posting on URL:

Header Name	Value	Description
x-digitap-callback-type	TRANSACTION_COMPLETE	Callback happens after transaction completed (successfully or otherwise)

Clients are advised to check the header data and then process the transaction complete data, as the same URL would be used to give other types of callbacks in the future.

The content type of the transaction complete callback would be “application/json” and below is the sample JSON structure:

```
{
  "txn_id": "daf4292",
  "status": "Success",
  "code": "",
  "message": "",
  "client_ref_num": "",
  "request_id": "",
  "periodic_consent_status": ""
}
```

Elements of the JSON are given below:

Elements	Description
txn_id	Digitap generated unique transaction id
status	Possible values are Success, Failure.
code	Code denoting the specific scenarios, a list is provided in each API section. Possible Value of codes
msg	Brief message description for either success or error response
client_ref_num	Client-specified identifier to uniquely identify this transaction. Alpha-numeric, or dash (-), maximum 50 characters.
request_id	Digitap provided unique request_id (<i>returned during the Generate URL API response</i>)
periodic_consent_status	ACTIVE, REVOKED, PAUSED, REJECTED, EXPIRED, PENDING

Notes on the response posted by Digitap on the callback URL.

- Digitap expects an HTTP 200 OK response from the URL on receipt of callback request
- If the first attempt fails, Digitap will try again two more times at regular intervals of 60 seconds.
- If a transaction concludes successfully, use the **Retrieve Report API** to download the generated report(s).
- If a transaction does not conclude successfully, use the **Status Check API** to find out more details about why.

B - Employer Details Parameter Specification

The Generate URL API takes the optional parameter **employer_details** which can be helpful in detecting the "salary" category for the transaction. This is the optional parameter, and would be useful only when the report type and subtype the client has opted for has the Transaction Categorization feature enabled. Please see Appendix [C - Report Types and Subtypes](#) for more details.

Sample Format :


```

employer_details - [
    {"name": "company 1 name",
    "CIN": "company 1 CIN number"},

    {"name": "company 2 name",
    "CIN": "company 2 CIN number"},
    ...
]

```

This parameter is a LIST of JSON elements consisting of two keys - name and cin.

- name - “Name of the Employer / Company”
- CIN- Company Identification Number Provided by the MCA

One of the elements - name or CIN is mandatory.

The **employer_details** parameter helps the Digitap to find the salary transaction more efficiently in the following ways:

- If the name is provided in the employer details section, then the transaction will be searched on the basis of the company name and will be categorized as a salary.
- If the CIN is provided, then the transaction will be searched on the basis of narration which Digitap has internally associated with the CIN and will be categorized as a salary. *(Digitap internally maintains the various Salary narration patterns for each of the companies and updates them on a periodic basis).*
- If both are provided then the category will be detected sequentially - first based on name and then based on CIN.

C - Report Types and Subtypes

Currently, Digitap provides a report (**report_type**) in two formats: JSON format and Excel format. Each format has different subtypes (**report_subtype**) and data provided in the report differs based on the subtype specified. The **report_type** and **report_subtype** can be specified while calling the Retrieve Report API.

Each client would be configured with allowed report types and report subtypes and default report type and subtype. Please contact the business team to understand more.

Basic Report Types and Subtypes are explained below:

- JSON Format - To get the report in json format, the report_type value should be passed as json. For the different subtypes value can be passed as follows:
 - a. **report_type** - "json"
report_subtype - "type1"
type1 - This denotes the simple report with raw data of transactions and user details.
 - b. **report_type** - "json"
report_subtype - "type2"
type2 - This denotes the simple report plus the categorisation added for each of the transactions.
 - c. **report_type** - "json"
report_subtype - "type3" or "type4" or others
type3 - This report contains the JSON data provided in Excel Type3

For other subtypes, please contact the business team
- Excel Format - To get the report in json format, the report_type value should be passed as xlsx.
 - a. **report_type** - "xlsx"
report_subtype - "type3"
type3 - Basic Excel Report type report which contains transactions list, daily EOD balances and summary variables.
 - b. **report_type** - "xlsx"
report_subtype - "type4" or other
type4 - This is the excel format for the json type3 report .

D - Note on Date Range

Clients can configure the default date range, which can be used for calculating start_month and end_month if not specified in the Generate URL API. Digitap supports the following date ranges

- a. **Full Months** - Last 'N' Full Calendar Months
- b. **Full Months+** - Last 'N' Full Calendar Months + Buffer Days of Current Month

The start_month and end_month for the above two cases is explained by the below example

- **Case 1 - Full Months**

Default Range	Date	Current Date	start_month	end_month	Remarks
6		18-Feb-21	Aug-2020	Jan-2021	Transactions from 1st Aug 2020 to 31 Jan 2021 will be considered.

- **Case 2 - Full Months+**

Default Range	Date	Current Date	start_month	end_month	Remarks
6+		18-Feb-21	Aug-2020	Feb-2021	Transactions from 1st Aug 2020 to 18 Feb 2021 will be considered.

If the client specifies **start_month** and **end_month**, then the start and end dates are calculated as follows:

- The start date of the statement will be the first day of the month using the **start_month** parameter.
- End date of the statement will be the least (min) of the last day of the month of **end_month** parameter, and current date.

E - Note on Acceptance Policies

If the client does not supply **acceptance_policy**, then the default acceptance policy value will be taken based on the client configuration.

- Acceptance policy considers months based on the **start_month** and **end_month** provided in the Generate URL API. If **start_month** and **end_month** are not specified, then it is taken based on the default date range (See Appendix [D - A note on Date Range](#) for more details)
- Acceptance examples for **atLeastOneTransactionInRange**

start month	end month	Current month	Transaction Available For	Transaction Available For Not	Accepted
Aug-2020	Jan-2021	Feb-2021	Feb	Aug-20 to Jan-21	No

Aug-2020	Feb-2021	Feb-2021	Feb	Aug-20 to Jan-21	Yes
Aug-2020	Jan-2021	Feb-2021	Aug, Sept	Oct-20 to Jan-21	Yes
Aug-2020	Jan-2021	Feb-2021	Jan	Aug-20 to Dec-20	Yes

- Acceptance examples for **atLeastOneTransactionPerMonthInRange**

start month	end month	Current month	Transaction Available For	Transaction Not Available For	Accepted
Aug-2020	Feb-2021	Feb-2021	Feb	Aug-20 to Jan-21	No
Aug-2020	Jan-2021	Feb-2021	Aug-20 to Jan-21	-	Yes
Aug-2020	Feb-2021	Feb-2021	Aug-20 to Jan-21	Feb-21	No
Aug-2020	Feb-2021	Feb-2021	Aug-20 to Feb-21	-	Yes

- Acceptance examples for **exactStatementRange**

start month	end month	Relax ation days	Statement start date	Statement end date	Current Date	Accept ed	Missing dates
Aug-2020	Feb-2021	0	01-Aug-20	18-Feb-21	18-Feb-21	Yes	
Aug-2020	Feb-2021	0	01-Aug-20	31-Jan-21	18-Feb-21	No	01-Feb-21 to 18-Feb-21
Aug-2020	Jan-2021	0	01-Aug-20	31-Jan-21	18-Feb-21	Yes	
Aug-2020	Jan-2021	0	01-Aug-20	31-Dec-21	18-Feb-21	No	01-Jan-21 to 31-Jan-21
Aug-2020	Jan-2021	10	01-Aug-20	10-Jan-21	18-Feb-21	Yes	

F - Return URL Integration

Digitap provide the facility to redirect to the client interface once transaction is complete and it can be done in following two ways:

1. Redirection to Return URL (*Suitable for clients using normal web pages as UI*)
2. Redirection to the Android Application Interface (*Suitable for clients have UI in Native Android*)

Redirection to Return URL:

If the return url is specified, Digitap sends the End User to this URL after completing interaction with Digitap.

Placeholders can be inserted in the URL to get a few data during the redirection. It supports two or three placeholders in the URL

1. **Two placeholders case** - will be replaced by the txn_id and true/false indicating Transaction success/failure in order
 - e.g. https://xxxui.client.com/process?txn_id=%s&status=%s
2. **Three placeholders case** - will be replaced by the request_id, txn_id, and true/false indicating Transaction success/failure in order
 - e.g. https://xxxui.client.com/process?request_id=%s&txn_id=%s&status=%s

User will be redirected to the Return URL once the transaction is complete, which will be one of the following cases:

- When User Clicked the Cancel Button in the Home Page
- When the User Clicked the Take me Back button in the Error Page
- Once Data Extraction completed (Success)

If the number of placeholders is not 2 or 3, then no replacement would occur.

Digitap offers two distinct redirection flows, providing clients the flexibility to choose based on their configuration:

1. **Redirection after Transaction Completion:** In this flow, after the entire transaction is successfully completed, Digitap will initiate a redirection to the designated return URL, till this time the user will be on Digitap Screen.
2. **Intermediate Redirection before Transaction Completion:** In this flow, once the user actions are completed i.e. No More User Interaction is required, Digitap will initiate a redirection to the designated return URL.
For example, in the **statementupload Flow**, once the user interaction is finished, Digitap will redirect the user to the return URL.

Redirection to the Android Application Interface:

If the client is using Native Android UI, then launch the Digitap UI in the webview with the JavaScriptInterface added with the DigitapBS Class as follows. Once the transaction is completed, Digitap Bank UI would invoke the onFinish() method of the DigitapBS class in which the client can close the webview and move back to the Native Android UI.

Digitap will pass the three parameters request_id, txn_id and status(true/false/in-progress) indicating Transaction success/failure during the onFinish() call

```
public class ClassName {  
  
    @JavascriptInterface  
    public void onFinish(String requestId, String txnId, String status) {  
        //Close the WebView  
    }  
}  
webView.addJavascriptInterface(new ClassName(), "DigitapBS");
```

Note that return_url should not be specified for this type of redirection.

Note that the status “in-progress” means that the transaction is not yet completed and client can use the code received in the webhook as the final status code of the transaction.

----- End of Document -----